Assignement 3

Q1. Vector magnitude

A = (2,3)
Magnitude = sqrt(13)
Concept: Norm

Q2. Vector addition

A+B = (4,6)
Concept: Vector addition

Q3. Scalar multiplication

4*(2,3)=(8,12)
Concept: Scalar multiplication

Q4. Dot product

1*3+2*4=11
Concept: Inner product

Q5. Cosine similarity

cosθ = dot/(||A|| ||B||)
Concept: Similarity

Q6. Orthogonal vectors

Dot=0 → orthogonal
Concept: Orthogonality

Q7. Projection

proj formula used
Concept: Projection

Q8. Linear independence

Rank check
Concept: Independence

CDAC Mumbai

Q9. Basis

RREF pivot columns
Concept: Basis

Q10. Linear combination

2A+3B
Concept: Combination

Q11. Span

All combinations
Concept: Span

Q12. Matrix addition

Element-wise
Concept: Matrix add

Q13. Matrix multiplication

Row×column
Concept: Multiplication

Q14. Transpose

Swap rows/columns
Concept: Transpose

Q15. Hadamard

Element-wise multiply
Concept: Hadamard

Q16. Determinant

Area scaling
Concept: Determinant

Q17. Identity matrix

I*A = A
Concept: Identity

CDAC Mumbai

Q18. Inverse

A^{-1} exists if det≠0
Concept: Inverse

Q19. Rank

Independent rows
Concept: Rank

Q20. Trace

Sum of diagonal
Concept: Trace

Q21. Symmetric

$A = A^T$
Concept: Symmetry

Q22. Orthogonal matrix

$A^TA = I$
Concept: Orthogonal

Q23. Eigenvalues

λ, v
Concept: Eigen

Q24. PCA

Dimensionality reduction
Concept: PCA

Q25. SVD

$A=USV^T$
Concept: SVD

Q26. Derivative

$d/dx(x^2+3x)=2x+3$
Concept: Derivative

CDAC Mumbai

Q27. Partial derivative

$\partial/\partial x$

Concept: Partial derivative

Q28. Gradient

Vector of derivatives

Concept: Gradient

Q29. Gradient descent

Iterative optimization

Concept: GD

Q30. Lagrange multipliers

Constraint optimization

Concept: Lagrange

CDAC Mumbai